

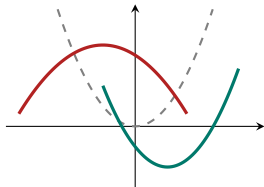
4-1 Quadratic Functions and Transformations

Prof. Fang Math

Experience first. Symbols second. Understanding always.

STUDENT NOTES

Name: _____



Today's Learning Targets

I can ...

- recognize the **parent** quadratic function and its graph
- describe how a , h , and k **transform** the graph
- find the vertex, axis of symmetry, max/min, domain and range from **vertex form**
- **write** a quadratic function from a graph or a description

Vocabulary

parabola	vertex form
vertex	minimum value
axis of symmetry	maximum value
reflection	translation
vertical stretch	vertical compression

Warm-Up: The Parent Function $f(x) = x^2$

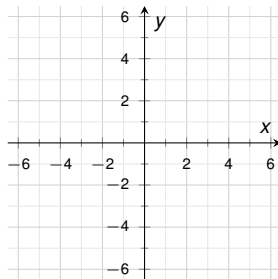
Complete the table, then plot the points.

x	-3	-2	-1	0	1	2	3
$f(x)$	<u> </u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>

Notice

What do you notice about the outputs for x and $-x$?

They are _____, so the graph is
_____.



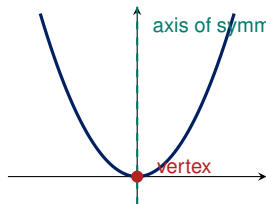
Definition: Quadratic Function

Key Idea

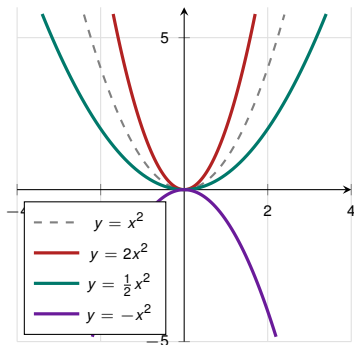
A **quadratic function** is a function that can be written as $f(x) = ax^2 + bx + c$, where _____.

Its graph is a U-shaped curve called a _____.

- The **vertex** is the _____ of the parabola.
- The **axis of symmetry** is the _____ through the vertex.
- For $f(x) = x^2$: vertex _____, axis _____



Explore 1: What Does a Do?



Notice

- $y = 2x^2$ is _____ than $y = x^2$.
(each y is _____)
- $y = \frac{1}{2}x^2$ is _____ than $y = x^2$.
- $y = -x^2$ opens _____ : it is a
_____ in the x -axis.
- The vertex stays at _____ every time.

Key Idea: The Graph of $f(x) = ax^2$

Value of a	Effect on the graph	Example
$ a > 1$	vertical _____ (narrower)	$y = 3x^2$
$0 < a < 1$	vertical _____ (wider)	$y = \frac{1}{4}x^2$
$a > 0$	opens _____; vertex is a _____	$y = 2x^2$
$a < 0$	opens _____; vertex is a _____	$y = -2x^2$

Common Mistake

“Stretch” and “compression” depend on $|a|$, not on the sign: $y = -3x^2$ is a reflection *and* a vertical _____.

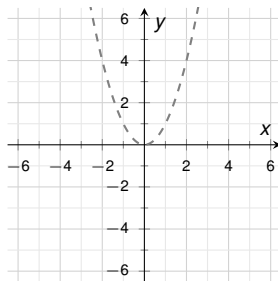
Watch Me 1: Graph $f(x) = -\frac{1}{2}x^2$

Watch Me

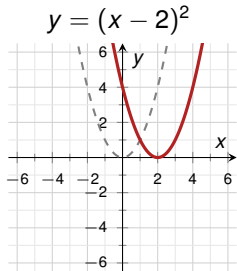
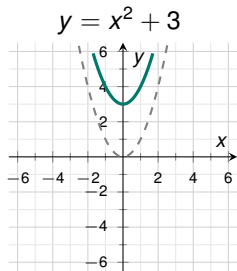
x	-4	-2	0	2	4
$f(x)$	<u> </u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>

Compared with $y = x^2$:

- reflection in the _____
- vertical _____ by a factor of _____
- vertex _____ is a _____



Explore 2: What Do h and k Do?



Notice

$y = x^2 + 3$ moves the graph
_____.

$y = (x - 2)^2$ moves the graph
_____.

Surprise? $x - 2$ *inside* the square moves the graph the _____ way of the sign.

Key Idea: Translations

Function	Translation of $y = x^2$	Vertex
$y = x^2 + k$	_____ if $k > 0$, _____ if $k < 0$	_____
$y = (x - h)^2$	_____ if $h > 0$, _____ if $h < 0$	_____
$y = (x + 3)^2$	_____ (here $h =$ _____)	_____

Common Mistake

$(x + 3)^2$ is $(x - (-3))^2$, so $h = -3$: the graph moves **left**, not right.

Putting It Together: Vertex Form

Vertex Form

$$f(x) = a(x - h)^2 + k$$

- vertex: _____
- axis of symmetry: _____
- $a > 0$: opens up, min value _____
- $a < 0$: opens down, max value _____
- domain: _____
- range ($a > 0$): _____
- range ($a < 0$): _____
- $|a|$ controls _____

Order to graph

_____ /stretch with $a \rightarrow$ shift _____ by $h \rightarrow$ shift _____ by k

Watch Me 2: $f(x) = (x + 3)^2 - 4$

Watch Me

Rewrite: $f(x) = 1(x - (\underline{\quad}))^2 + (\underline{\quad})$

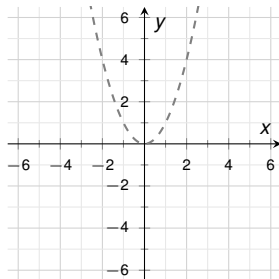
$a = \underline{\quad}$, $h = \underline{\quad}$, $k = \underline{\quad}$

Transformations: $\underline{\quad}$ 3, $\underline{\quad}$ 4

Vertex $\underline{\quad}$ Axis $\underline{\quad}$

$\underline{\quad}$ value: $\underline{\quad}$

Domain: $\underline{\quad}$ Range: $\underline{\quad}$



Watch Me 3: $f(x) = -2(x - 1)^2 + 5$

Watch Me

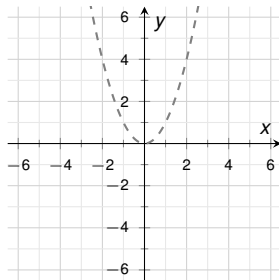
$a =$ _____, $h =$ _____, $k =$ _____

Transformations (in order):

_____, vertical _____ by _____,
_____ 1, _____ 5

Vertex _____ Axis _____

_____ value: _____ Range: _____

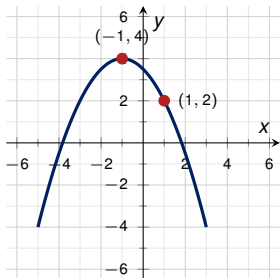


Your Turn: Read the Vertex Form

Your Turn

Function	Vertex	Axis	Max or Min?	Value
$f(x) = (x - 5)^2 + 2$	_____	_____	_____	_____
$g(x) = -(x + 1)^2 - 3$	_____	_____	_____	_____
$p(x) = \frac{1}{2}x^2 - 6$	_____	_____	_____	_____

Watch Me 4: Write the Equation from a Graph



Watch Me

Step 1. Vertex $(h, k) =$ _____
 $f(x) = a(x + \underline{\hspace{1cm}})^2 + \underline{\hspace{1cm}}$

Step 2. Use another point $(1, 2)$ to find a :
 $2 = a(1 + 1)^2 + 4$
 $2 = \underline{\hspace{1cm}} \Rightarrow a = \underline{\hspace{1cm}}$

Step 3. $f(x) = \underline{\hspace{2cm}}$

Check: opens _____, so a should be _____
✓

Watch Me 5: Write the Equation from a Description

Watch Me

Start with $y = x^2$. Reflect it in the x -axis, stretch it vertically by a factor of 3, then translate it **4 units left** and **1 unit down**.

reflect + stretch by 3 $\Rightarrow$ $a =$ _____

4 units left $\Rightarrow$ $h =$ _____

1 unit down $\Rightarrow$ $k =$ _____

$g(x) =$ _____

Common Mistake

Left 4 means $h = -4$, which appears as $(x + 4)$ in the equation.

What If ... ?

What If?

1. What if a changes from 2 to -2 — does the vertex move?

2. What if you swap h and k : $(x - 2)^2 + 5$ vs. $(x - 5)^2 + 2$?

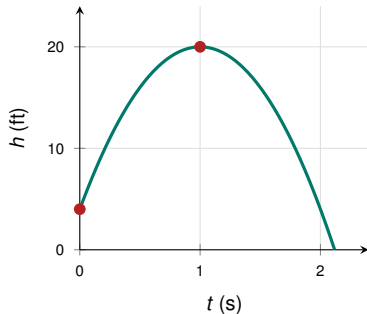
3. What if $a = 0$? Is $y = 0(x - 1)^2 + 3$ still a parabola?

Connection: A Water Fountain

The height (feet) of a water stream t seconds after it leaves the fountain is

$$h(t) = -16(t - 1)^2 + 20.$$

1. Maximum height: _____
2. It reaches the top after _____.
3. Starting height $h(0) =$ _____
4. Why is the vertex a maximum?



Your Turn

1. Describe the transformations of $y = x^2$: $g(x) = 3(x - 2)^2 + 1$

2. Describe the transformations of $y = x^2$: $g(x) = -\frac{1}{4}(x + 5)^2$

3. Write the function: $y = x^2$ reflected in the x -axis, then 6 right and 2 up.

$g(x) =$ _____

4. Vertex $(3, 1)$; the graph passes through $(5, 9)$. $f(x) =$ _____

Challenge

A parabola $f(x) = a(x - h)^2 + k$ passes through $(0, 5)$ and $(4, 5)$, and its minimum value is -3 . Find $f(x)$.

Same height at $x = 0$ and $x = 4 \Rightarrow$ axis of symmetry $x = \underline{\hspace{2cm}}$

Minimum value $-3 \Rightarrow$ vertex $\underline{\hspace{2cm}}$

$$5 = a(0 - 2)^2 - 3 \Rightarrow a = \underline{\hspace{2cm}}$$

$$f(x) = \underline{\hspace{4cm}}$$

Summary

Big Picture

Parent function

$y = x^2$, vertex _____, axis _____

Vertex form

$f(x) = a(x - h)^2 + k$, vertex _____

a

sign: opens up/down; $|a|$: _____

h

horizontal shift — _____ of the sign in $(x - h)$

k

vertical shift; also the _____ value

Exit Ticket

For $f(x) = -3(x + 2)^2 + 7$:

1. Vertex: _____ Axis of symmetry: _____
2. Opens _____, so 7 is the _____ value. Range: _____
3. Transformations of $y = x^2$:

Homework: practice set 4-1 · Extra practice: Math Practice Generator, Chapter 4 → Quadratic Transformations